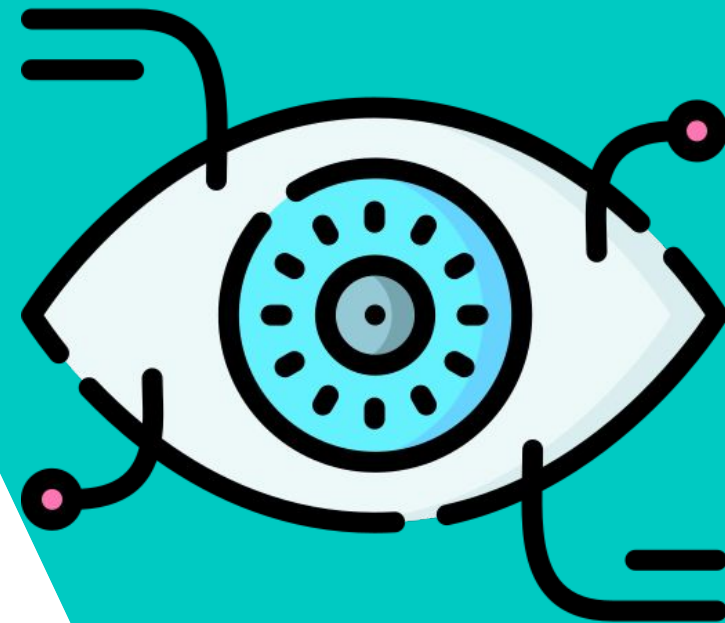


Computer Vision

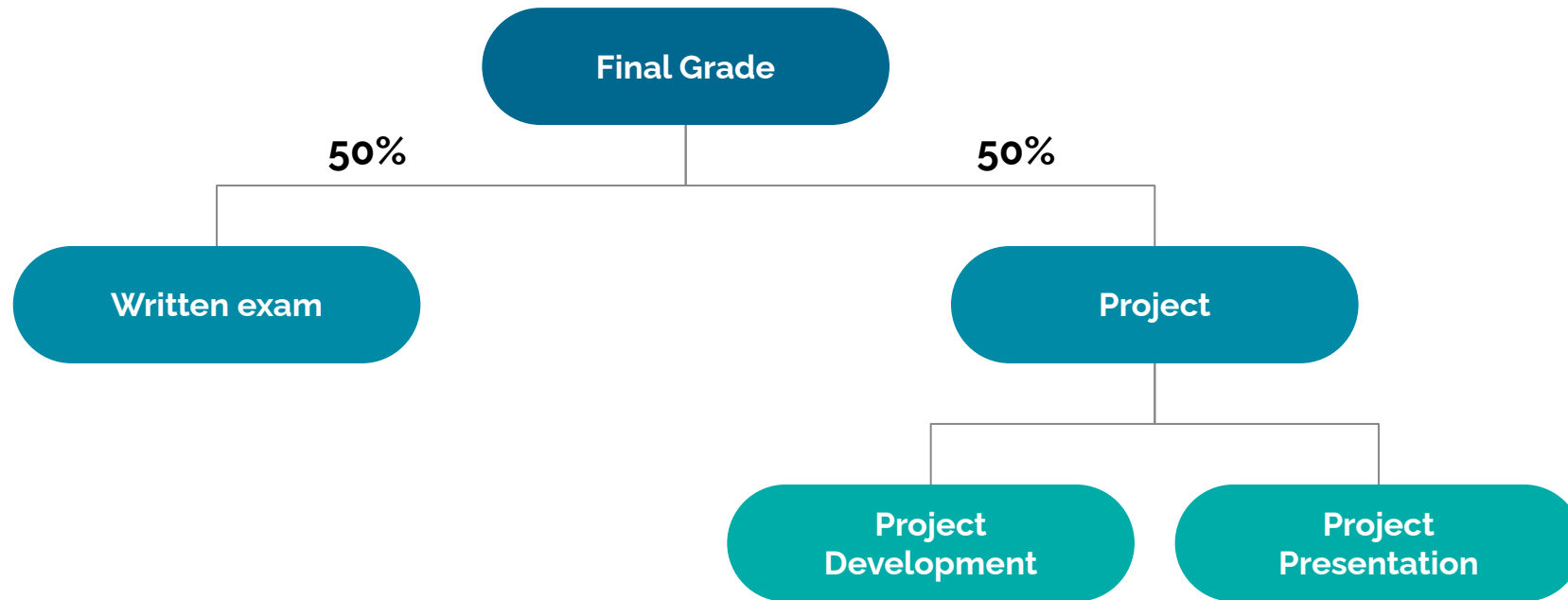
A.A. 2025-2026
Exam guidelines



SAPIENZA
UNIVERSITÀ DI ROMA



Exam structure



Exam registration

- The written exam and project can be taken at **different times**
 - We will keep your grade for each part
 - Final grade will be registered once you have completed both parts
 - Written exam and project must be completed within the **academic year i.e., between the June 2026 session and the March 2027 session.**

* October and March session are reserved to "categories of students referred to in Article 21, paragraph 7, of the General Study Manifesto
<https://www.uniroma1.it/it/content/esami-di-profitto-idoneita-tirocini-altre-attivita-formative>

- If you complete only one part, a refused will be recorded
 - This is **not a negative thing**, it's just the standard procedure
 - We will keep the grade for the part you have completed
- A Google Form will be provided to communicate your project choose

Exam registration

3 possible cases:

- 1) **You take the written exam and project presentation in the same session**
 - register on infostud
 - fill in the project Google Form
- 2) **You take first the written exam in a specific session and the project presentation in another session**
 - register on infostud for the written exam
 - a refused/rinuncia will be recorded (*just to keep your grade, don't worry!!*)
 - fill in the project Google Form for the project presentation and register on infostud when taking the project part
- 3) **You take first the project presentation in a specific session and the written exam in another session:**
 - fill in the project Google Form when taking the project presentation
 - register on infostud when taking the written exam

Project guidelines

- **Objective:** develop a non-trivial solution to a **proposed** topic
 - Develop **all the objectives** of your chosen project
 - Show critical thinking and problem solving skills
 - Introduce small novelty in the approach, feel free to experiment
- This is a **hands-on project**, not a survey
 - Focus on methodology and approach
- The project can be carried out by a **group**
 - Maximum **3 members**
 - Projects can also be completed by individual students.
 - *We suggest to work in team*
- **Plagiarism will be strictly penalized**

OPEN Project guidelines

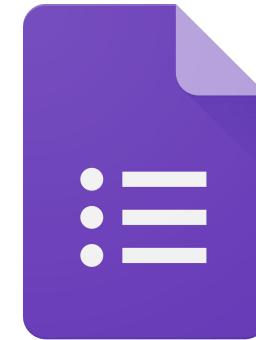
- All the previous guidelines apply! (Slide 5)
- **Proposal:** Students choosing this project must submit
 - A topic proposal summarizing their chosen topic (max 1 page)
 - At least two reference papers, underlining their current limitations
 - Highlight the novelty of the proposal
- Students must select a **backup project among our proposed topics**, in case their proposal gets rejected
- Choosing the OPEN Project is exclusive to June-July session!

How to Communicate the Project Topic

- All the students willingly to deliver the project in the **June-July session**, must fill this Google Form by **May 10th**, in order to select the preferred project topic.

For students choosing the OPEN Project, acceptance will be notified by **May 15th**.

Only one form registration per group is needed.



Form Link

make sure to login with the
@[studenti.uniroma1.it](#) account

Code guidelines

- **Deep Learning framework:** PyTorch
 - NO exceptions
- **Code:** Google Colab
 - Or other notebooks (Kaggle, etc.)
 - You can also VSCode in local or similar
- In all cases, you **MUST** follow a precise code structure
 - Try to maintain as much as possible the conceptual structure
- **No tutoring** for the project development

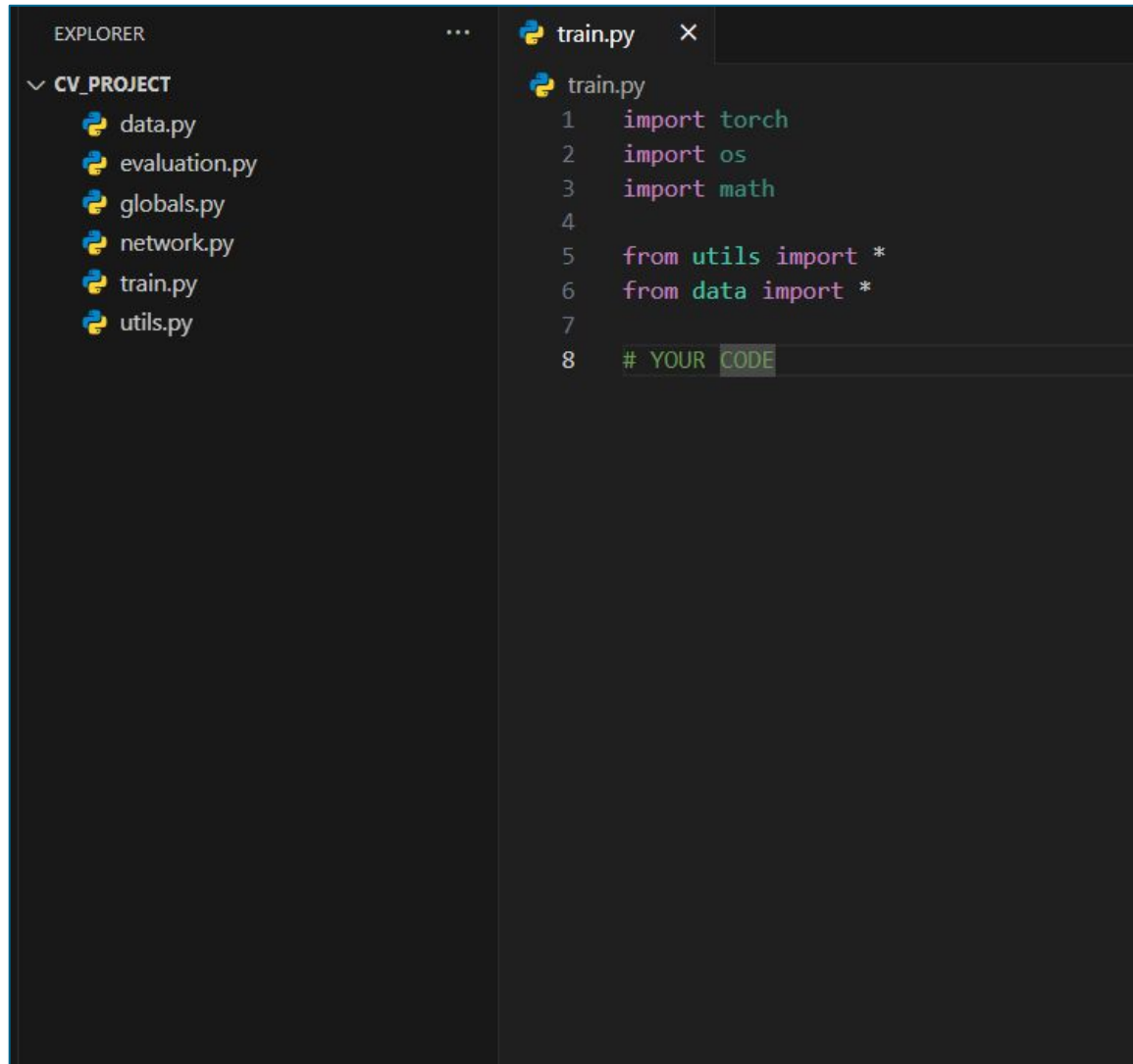
Code structure

- **Imports:** all the needed packages
 - This section is for the notebook projects
- **Globals:** useful variables on the whole code
- **Utils:** code support functions
- **Data:** everything related to data management
- **Network:** code to structure the neural network
- **Train:** part containing the training cycle elements
- **Evaluation:** tests needed for the trained network

Imports
[] <code>import torch</code>
Globals
[] <code>batch_size = 8</code> <code>lr = 1e-5</code> <code>...</code>
Utils
[] <code>def everywhere(param):</code> <code> result = f(param)</code> <code> return result</code>
Data
[] <code>dataset = MyDataset()</code>
Network
[] <code>model = MyModel()</code>
Train
[] <code>model.train()</code>
Test
[] <code>model.eval()</code>

Code structure

Notebook-like example



The image shows a screenshot of a code editor interface, likely VSCode, with a dark theme. On the left, the 'EXPLORER' sidebar shows a project named 'CV_PROJECT' containing several Python files: data.py, evaluation.py, globals.py, network.py, train.py, and utils.py. The 'train.py' file is selected and open in the main editor. The code in 'train.py' includes imports for torch, os, and math, followed by wildcard imports from 'utils' and 'data' modules. The last line is a comment placeholder: '# YOUR CODE'.

```
train.py
1  import torch
2  import os
3  import math
4
5  from utils import *
6  from data import *
7
8  # YOUR CODE
```

Code structure

VSCoDe-like example

When to Register Your project for the final Exam

- A list of projects will be published according this schedule:
 - **June-July** → list available by the end of April
 - **September-October** → list available by the end of July
 - **Other sessions** → list available by the end of October
- It is **not** allowed to submit a project outside the session for which it was proposed.
 - **No exceptions** will be made

How to Register for the Project Presentation

- **Google Form (to deliver 7 days before the exam date)** in which specify
 - Group members
 - Name, Surname, Student-ID, mail
 - Project chosen
 - GitHub repository link
- **GitHub** repository should contains
 - Code
 - Dataset (or the link to it)
 - Project presentation
 - Detailed Readme

this google form will be published in proximity of the exam date

How to Register for the Project Presentation

- **Please remember that**
 - The form must be submitted **no later than 7 days** before the date on which you intend to present the project
 - The repository can be modified **up to 2 days** before the date on which you intend to present the project

Presentation guidelines

- Upload the presentation to **GitHub** and saved on a **USB** drive
 - You can use PDF, PowerPoint, or both formats.
- Use **white background** slides
- Use **high color contrast** elements for plots and images
- Each presentation should last **10 minutes**
- Avoid jumping forward and backward during the presentation
- Ensure to include page numbers on each slide
 - Show the current slide number out of the total slide number

Presentation suggested scheme

- **Title:** Report the project name, your names, and course information
- **Outline:** Briefly describe the presentation workflow
- **Problem Statement:** The challenge you are addressing
- **State of the Art:** How current research approaches this challenge
- **Proposed Method:** How you approached this challenge
- **Dataset:** The data you used for the project
- **Experimental Setup:** How you configured the elements of the project
- **Model Evaluation:** How you assessed the performance of your model
- **Conclusions:** Final considerations and future work
- **References:** Where you drew inspiration from

What is the purpose of all these rules?

- These rules aim to instill in you a **professional approach**
- We **firmly believe** that this will benefit you in all your future tests
 - This is a standardized method to present **any project**
- Don't be intimidated by the challenges
 - The important thing is to adopt an engineering mindset
 - From this, we focus on the methods you propose
 - And how you present your ideas

Exam dates



Next appointments



Title: Self-supervised learning, JEPA, world models, and AI's future

Alfredo Canziani: Assistant Professor of Computer Science at Courant Institute of Mathematical Sciences, mentored by professor Yann LeCun.

21/05/2026, DIAG B2, Online & In presence



Title: AI and Computer Vision for Guidance, Navigation, and Control of Deep-Space Robotic Systems

Antonio Genova: Associate Professor in the Department of Mechanical and Aerospace Engineering (DIMA) at Sapienza, leading the SPace Robotics

INvestigation Group (SPRING).

27/05/2026, DIAG B2, In presence

Projects List

June-July session